



Source apportionment and controlling mechanisms of nitrogen leaching from bioretention systems

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ABSTRACT

Nitrogen (N) leaching has attracted concerns in bioretention systems (BRSs), while regulation of N leaching in BRSs is a challenge. In this study, N leached from BRSs with three plants (*Phragmites australis*, *Acorus calamus* L. and *Cana indica* L.) has for the first time been apportioned between three types of sources: immediate, fast and slow leaching based on ¹⁵N labeling. Results indicated that: fast leaching mainly comes from the previous four rainfall-runoffs, while slow leaching accounts for the largest proportion in total leaching; immediate leaching is mainly as NO₃⁻, fast leaching mainly as NO₃⁻ and dissolved organic nitrogen (DON), and slow leaching mainly as DON and NH₄⁺; BRSs with different plants vary in source compositions of N leaching. Meanwhile, N fates, transformation rates, and functional genes were constrained to explain the controlling mechanisms of each source of N leaching: immediate leaching mainly relates to soil characteristics and root morphology (e.g. type, surface area and diameter) that can influence adsorption, preferential flow and plant uptake; fast leaching mainly relates to soil N transformation during the dry periods that can influence the availabilities of NO₃⁻ and DON; slow leaching mainly relates to plant uptake and soil retention, soil N transformation and plant-soil interactions can influence the availabilities and stability of DON and NH₄⁺. This study provides insights into the source compositions of N leaching, the long-term N legacy effect and controlling mechanisms of N leaching in BRSs-like engineering systems, which will imply future improvements in design and risk assessment of green infrastructure.

1. Introduction

Bioretention systems (BRSs), also referred as rain gardens, are generally composed of vegetation, soil media, sand layer, gravel layer and a drainage component [8]. As widely-used low impact development (LID) practices for urban runoff management, BRSs can effectively reduce metals, suspended sediments and pathogens in urban runoff [9, 33]. However, nitrogen (N) leaching always occurred in BRSs, leading to unstable N removal efficiency [22]. This problem restricts the popularization and application of BRSs. Hence, there are pressing needs to address N leaching problems of BRSs.

N leaching in BRSs could potentially be attributed to complex sources. A large number of studies showed that some N species such as NO₃⁻ are not easily adsorbed and can directly leach out during the current rainfall-runoff [12,27,29]. Meanwhile, N from recent rainfall-runoffs

may first get withheld by soil particles, and then leach shortly after temporary storage [32,35]. Furthermore, part of the inner N pool originated from long-ago rainfall-runoffs, which is generally considered “stable and inert”, may still leach. Therefore, N leached from BRSs may come from the current rainfall-runoff (immediate leaching), the recent rainfall-runoffs (fast leaching) and long-ago rainfall-runoffs (slow leaching). The three entangled sources are gaining awareness but have not been well-differentiated, making the real reasons for N leaching ambiguous.

Actually, immediate leaching is widely recognized by engineers, while fast and slow leaching are usually ignored. Plenty of studies have monitored the rainfall-runoff events in BRSs to find immediate leaching of significant N as a result of high N loads in influents, which might relate to soil physical properties and plant types [16,18,31]. However, there were also occasional leaching of high loads of N that could not be

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explained by the influents, but suggest fast or slow leaching of N legacy from previous rainfall-runoff events [4]. Few studies have systematically studied the fast or slow leaching in BRSs, not to mention the controlling mechanisms, though they are insightful for the long-term N release risks and future improvements in BRS design [23].

^{15}N labeling technique has proved powerful in constraining N cycles in various ecosystems, with emerging applications in BRSs. Some studies have traced the fates of influent N in soil, plant, effluent and gas emissions based on ^{15}N labeling experiments [6,23,30], while a few studies have measured N transformation rates by incubation experiments in BRSs [2,7,21,30]. Further, several studies tried to account for N leaching by the retention by soil or plant or transformation rates, but not always as expected. This could also be attributed to the failure to consider fast and slow leaching. However, due to the specificity and traceability, ^{15}N labeling technique could still be instrumental to constrain the source compositions of N leaching and understand the connections between N transport or transformations and each source of N leaching in BRSs.

This study proposes the presence of three sources of N leaching from BRSs, i.e. immediate leaching, fast leaching and slow leaching. Then, a methodology is for the first time developed to apportion the three sources by advantage of ^{15}N labeling technique, which has been exemplified in nine BRSs with three different plant species under greenhouse conditions. Finally, the comprehensive controlling mechanisms of each source of N leaching are specifically unraveled, and the implications and limitations of this study are further discussed.

2. Materials and methods

2.1. Scheme of three sources in N leaching of BRSs

Effluent N in each rainfall-runoff event is proposed to come from the following three sources: 1) Immediate leaching: with N sourced from the current rainfall-runoff event; 2) Fast leaching: with N sourced from recent rainfall-runoff events after a short stay; 3) Slow leaching: with N sourced from long-ago rainfall-runoff events after a relatively long-period stay. The schematic diagram is shown in Fig. 1. These three leaching sources are entangled all the time [4,30], but can be distinguished based on our developed methodology by advantage of ^{15}N labeling technique.

2.2. Experimental setup and ^{15}N tracing experiment

To exemplify the source apportionment of N leached from BRSs between immediate leaching, fast leaching and slow leaching, nine experimental columns were designed to conduct ^{15}N tracing experiment.

The columns used in the ^{15}N tracing experiment were made of acrylic material (diameter: 50 cm). Each column has a 30 cm thick saturated zone (including 10 cm thick gravel layer and 20 cm thick sand filter layer), 50 cm thick soil medium layer, and vegetation (Fig. 2), with more details described in Wang's study [30].

Three plant species that are flood- and drought-tolerant, i.e. *Canna indica* L., *Phragmites australis*, *Acorus calamus* L., were selected. Each system was planted with a single-species community of plants and recorded as BR-C, BR-P or BR-A, with three parallel systems for each plant species. The plants were grown into seedlings (approximately 50 cm height), then transplanted into the BRSs (12 plants per column), and each system was stabilized with tap water once every three days in the first two weeks to make soil sink and plant root. All the columns were placed in a transparent canopy at Peking University Shenzhen Graduate School, Shenzhen, China.

Before the ^{15}N tracing experiment, non-labeled rainfall-runoff was introduced into the BRSs to acclimate for one month, whose composition was prepared as follows: NH_4Cl (2.5 mg $\text{N}\cdot\text{L}^{-1}$), KNO_3 (2.5 mg $\text{N}\cdot\text{L}^{-1}$), KH_2PO_4 (1 mg $\text{P}\cdot\text{L}^{-1}$), $\text{C}_3\text{H}_7\text{NO}_2$ (5 mg $\text{N}\cdot\text{L}^{-1}$), $\text{C}_6\text{H}_{12}\text{O}_6$ (200 mg $\text{C}\cdot\text{L}^{-1}$). The volume of added rainfall-runoff was set to be 70 L by sizing the systems to 5% of its impervious catchment area, corresponding to

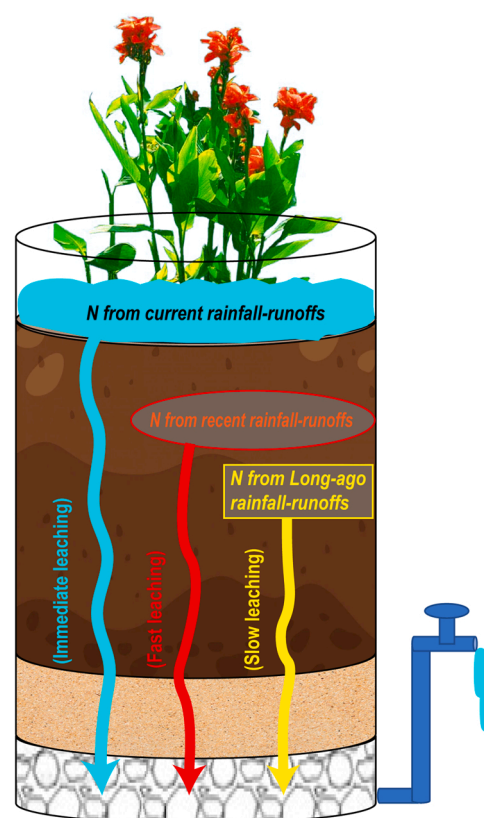


Fig. 1. Schematic diagram of source apportionment of N leached from BRSs.

rainfall intensity of 20 mm/h, runoff coefficient of 0.9 and rain event duration of 1 h. The antecedent dry period (ADP) was set at three days to conform with the average rainfall event interval in rainy season of Shenzhen city. The mimicked rainfall-runoff was applied using a pump (ML220, China) with an inflow rate of $1.2 \text{ L}\cdot\text{min}^{-1}$. The effluent of each rainfall-runoff event was monitored to make sure each BRS has eventually achieved a steady N removal efficiency.

After acclimation, the nine BRSs were one-time added with 70 L ^{15}N -labeled influent ($^{15}\text{NH}_4\text{Cl}$: 99.0 atom%; K^{15}NO_3 : 99.0 atom% and ^{15}N -urea: 10.0 atom%, the same concentrations as for acclimation) to mimic one rainfall-runoff event. The following four rainfall-runoff events were then mimicked using non-labeled rainfall-runoff and monitored of N (including ^{15}N) leaching. The mimicked non-labeled rainfall-runoff was also the same as for acclimation. The effluent for each rainfall-runoff event was collected, recorded of the total amount, and frozen before analyzing the contents of various N species and ^{15}N .

2.3. Principles of source apportionment

The three sources of N leaching can potentially be distinguished based on ^{15}N labeling techniques. Assuming the relative contributions of immediate leaching, fast leaching and slow leaching are consistent in each effluent after BRSs achieved steady conditions, immediate leaching can be determined based on the ^{15}N recovery rate in the first rainfall-runoff event; fast leaching can be determined based on the sum of ^{15}N recovery rates in the second till M th rainfall-runoff events during which ^{15}N content approaches to natural abundance; slow leaching can be considered as the difference between the total N in effluent and the sum of immediate leaching and fast leaching. The approximation of effluent ^{15}N content to natural abundance signals that leaching from the ^{15}N labeled rainfall-runoff event continues but becomes less and insignificant [30], which, however, could cumulate with those from other rainfall-runoff events to account for significant slow leaching. Therefore,

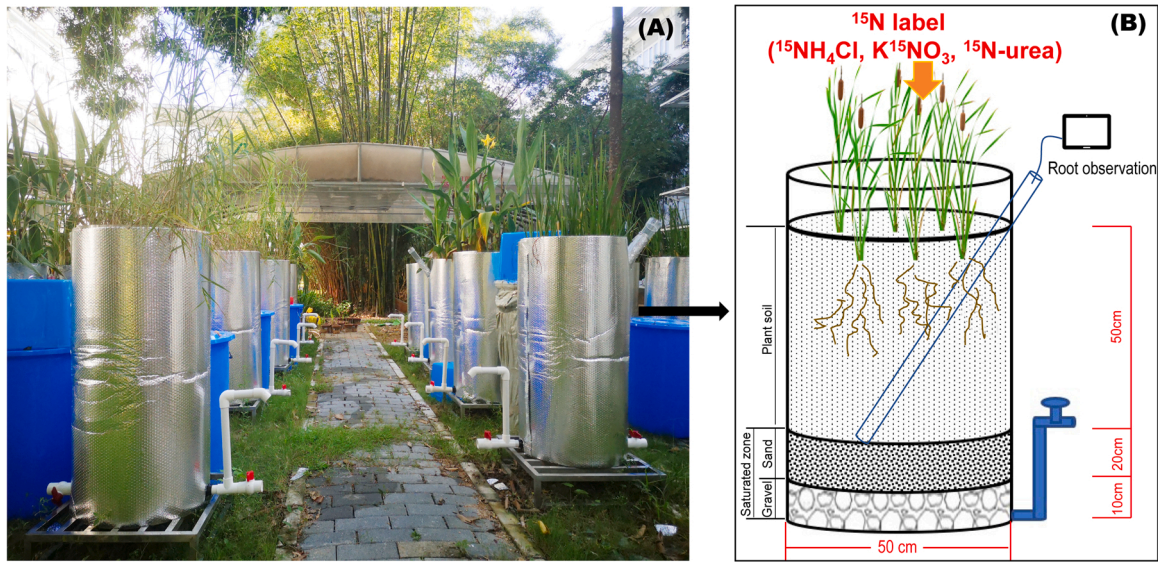


Fig. 2. Picture (A) and scheme (B) of BRSs.

the effluent ^{15}N content can help distinguish between fast leaching and slow leaching. The contributions of immediate leaching, fast leaching and slow leaching can then be derived as follows:

$$\text{Immediate leaching}(\%) = \frac{C_1 \times V_1 \times AT_1\%}{N} \times \frac{L_{in}}{L_{out}} \times 100\% \quad (1)$$

$$\text{Fast leaching}(\%) = \sum_2^M \frac{C_i \times V_i \times AT_i\%}{^{15}\text{N}_{in}} \times \frac{L_{in}}{L_{out}} \times 100\% \quad (2)$$

$$\text{Slow leaching}(\%) = \left(1 - \sum_1^M \frac{C_i \times V_i \times AT_i\%}{^{15}\text{N}_{in}}\right) \times \frac{L_{in}}{L_{out}} \times 100\% \quad (3)$$

where C_i ($i = 1 \sim M$) are the total nitrogen (TN) concentration of effluent in the i th rainfall-runoff event, respectively ($\text{mg N} \cdot \text{L}^{-1}$); V_i are the effluent volume in the i th rainfall-runoff event, respectively (L); $AT_i\%$ are the atom % excess of ^{15}N -TN (^{15}N in total nitrogen) in effluent in the i th rainfall-runoff event, respectively; $^{15}\text{N}_{in}$ is the mass of influent ^{15}N (mg); M is the serial number of the last rain where ^{15}N content approaches to natural abundance. L_{in} , L_{out} are the steady-state average TN loads in the influent and effluent (g), respectively.

Proportions of N species are calculated as follows:

$$\text{P}^{15}\text{N}_S(\%) = \frac{C_{s,i} \times AT_{s,i}\%}{C_i \times \alpha \times AT_i\%} \times 100\% \quad (4)$$

$$\text{PN}_S, \text{IL}(\%) = \frac{C_{s,1} \times AT_{s,1}\%}{C_1 \times \alpha \times AT_1\%} \times 100\% \quad (5)$$

$$\text{PN}_S, \text{FL}(\%) = \frac{\sum_2^M (C_{s,i} \times V_i \times AT_{s,i}\%)}{\sum_2^M (C_i \times V_i \times \alpha \times AT_i\%)} \times 100\% \quad (6)$$

$$\text{PN}_S, \text{SL}(\%) = \frac{1 - \sum_1^M (C_{s,i} \times V_i \times AT_{s,i}\% / N_{in}^{15})}{1 - \sum_1^M (C_i \times V_i \times \alpha \times AT_i\% / N_{in}^{15})} \times 100\% \quad (7)$$

where P^{15}N_S is the proportion of N species (S denotes the N species, i.e. NH_4^+ , NO_3^- and DON) to TN originated from ^{15}N -labeled influent in the i th rainfall-runoff event ($i = 1 \sim M$); PN_S, IL , PN_S, FL , PN_S, SL are proportions of N species to TN in immediate leaching, fast leaching and slow leaching, respectively; $C_{s,i}$ are N species concentrations of outflow in the i th rainfall-runoff ($\text{mg N} \cdot \text{L}^{-1}$), respectively; $AT_{s,i}\%$ are the atom% of ^{15}N

species of outflow in the i th rainfall-runoff event, respectively; α is the correction factor of atom% excess of ^{15}N -TN (in this study, α is 1.89 due to that ^{15}N -urea is only 10 atom%).

2.4. Controlling mechanisms of each source of N leaching

To explain the controlling mechanisms of each source of N leaching in BRSs with different plant species, N transformation rates, fates of ^{15}N -labeled influent, root morphology and abundances of functional genes were further investigated.

(1) N transformation rates.

In each experimental column, the soil (0–60 cm depth) was sampled during the dry period using a core sampler with 2 cm diameter for five replicates, and the collected soil samples were evenly mixed before influent ^{15}N labeling experiment. Each soil media sample was sieved to 2 mm mesh size with fine roots manually removed. One subsample (4.0 g dry soil) was added with suitable deionized water and adjusted to the actual moisture contents of BRSs, then the mixture was homogeneously labelled by either $^{15}\text{NH}_4\text{Cl}$ (99 atom%) or K^{15}NO_3 (99 atom%) with the negligible increases in ^{15}N abundance of both soil NH_4^+ and NO_3^- pools by $1.1 \text{ mg N} \cdot (\text{kg soil})^{-1}$. After 24 h incubation, gross N transformation rates were determined by ^{15}N isotope pool dilution method [19], the net N transformation rates were calculated by variations in mineral N concentrations over time. The calculations are shown as follows:

$$\text{GM} = \frac{[\text{NH}_4^+]_1 - [\text{NH}_4^+]_2}{t_2 - t_1} \times \frac{\log(\text{APE}_1 / \text{APE}_2)}{\log([\text{NH}_4^+]_1 / [\text{NH}_4^+]_2)} \quad (8)$$

$$\text{GN} = \frac{[\text{NO}_3^-]_1 - [\text{NO}_3^-]_2}{t_2 - t_1} \times \frac{\log(\text{ape}_1 / \text{ape}_2)}{\log([\text{NO}_3^-]_1 / [\text{NO}_3^-]_2)} \quad (9)$$

$$\text{NM} = \frac{[\text{NH}_4^+]_2 - [\text{NH}_4^+]_1}{t_2 - t_1} \quad (10)$$

$$\text{NN} = \frac{[\text{NO}_3^-]_2 - [\text{NO}_3^-]_1}{t_2 - t_1} \quad (11)$$

$$\text{I}_a = \text{GM} - \text{GN} - \text{NM} \quad (12)$$

where GM, GN, NM, NN, I_a are rates of gross mineralization, gross nitrification, net mineralization, net nitrification and NH_4^+ immobilization rate ($\text{mg N} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$); $[\text{NH}_4^+]_1$ and $[\text{NH}_4^+]_2$ are the concentrations of

NH_4^+ at time t_1 and t_2 , respectively ($\text{mg N}\cdot\text{L}^{-1}$); $[\text{NO}_3]_1$ and $[\text{NO}_3]_2$ are the concentrations of NO_3^- at time t_1 and t_2 , respectively ($\text{mg N}\cdot\text{L}^{-1}$); APE_1 , APE_2 are the atom% excess of $^{15}\text{N}\text{-NH}_4^+$ corresponding to the time t_1 and t_2 , respectively; ape_1 , ape_2 are the atom% excess of $^{15}\text{N}\text{-NO}_3^-$ corresponding to the time t_1 and t_2 , respectively.

(2) ^{15}N fates.

To explore the fates of labeled ^{15}N in BRSs, soil and plant samples were collected in each column after three months. The soil samples were collected in layers (0–20, 20–40 and 40–60 cm) using a core sampler with 2 cm diameter, before analysis. The samples were dried naturally, passed through a 2 mm mesh sieve and picked out plant roots. The plant samples were first separated into four parts: roots, stems, live leaves and withered leaves. Before N analysis, all samples were then oven dried at 65 °C and crushed. ^{15}N accumulation percentages by soil and plant tissues are calculated as:

$$D_j = \frac{\text{APE}_j \times T_j \times M_j}{^{15}\text{N}_{in}} \times 100\% \quad (13)$$

where D_j is the ^{15}N accumulation percentages by soil and plant tissues, j denotes soil or plant tissues; APE_j is the atom% excess $^{15}\text{N}\text{-TN}$ in j ; T_j is the TN contents in j (wt% in plant tissues or soil); M_j is the mass (g, dry weight).

The gaseous ^{15}N loss mainly due to denitrification process (L_N , mg) can be solved according to the ^{15}N mass balance as follows:

$$L_N = I_N - (E_N + M_N + P_N) \quad (14)$$

where I_N is the influent ^{15}N load (mg), E_N is the effluent ^{15}N load (mg), M_N is the ^{15}N captured by media (mg), P_N is the ^{15}N accumulated in plant biomass (mg).

(3) Root morphology.

To measure the root morphological parameters of different plant species, AZR-100 root ecosystem detection system (Aozuo instrumentation Ltd., Beijing, China) was installed in three experimental columns. The roots of three plant species were scanned and taken photos during the dry period. Root morphological parameters (e.g. mean diameter and root surface area) were analyzed using the root analysis software of WinRHIZO Tron MF.

(4) Abundances of functional genes.

Total DNA of each fresh soil sample collected during the dry period was extracted using the FastDNA SPIN Kit (MoBio Laboratories, Carlsbad, CA) according to the manufacturers' instructions. Quantitative polymerase chain reaction (qPCR) assays were performed on ABI 7300 Real-time PCR system (Applied Biosystems, America) for functional genes *AOA amoA*, *AOB amoA*, *nirK*, *narG* and *nosZ* using the primers given in Table S1. The PCR program were set as follows: initial denaturation for 3 min at 95 °C, then 40 cycles at 95 °C for 5 s, annealing at 58 °C for 30 s, and extension at 72 °C for 1 min. The amplification efficiencies ranged between 86% and 108%.

2.5. Chemical and isotopic analysis

The collected inflow and outflow samples were first filtered with 0.45 μm cellulose acetate membrane and then determined of NH_4^+ , NO_3^- and NO_2^- concentrations by Discrete Chemistry Analyzer (CleverChem200, Germany) based on the methods of salicylic acid spectrophotometry (HJ 536–2009), the hydrazine sulfate-NEDD spectrophotometry (GB/T5750.5–2006), and naphthalene ethylenediamine hydrochloride spectrophotometry (GB/7493–87), respectively. TN of the inflow and outflow was first digested with alkaline potassium persulfate to be completely converted to NO_3^- for measurement as NO_3^- (HJ 636–2012). DON concentrations were calculated by subtracting NO_3^- , NH_4^+ and NO_2^- from TN concentrations. The NO_2^- concentrations were all lower than the detection limit and not further discussed in this paper.

The soil moisture was determined by gravimetric method. Soil

samples were extracted of NO_3^- and NH_4^+ using 20.0 mL of 2 M KCl (1:5 mass to volume), and the slurry was then centrifuged for 10 min at 8000 rpm, finally filtered through 0.22 μm fiber filters. The NH_4^+ and NO_3^- concentrations in the filtered soil extracts were determined by Discrete Chemistry Analyzer as in water samples. Oxidation-reduction potential (ORP) in two soil layers (upper: 0–20 cm; lower: 20–40 cm) and saturated layer were measured during the dry period, by ORP meter (MTC30, Hash, America).

The TN contents and $^{15}\text{N}\text{-TN}$ abundance in the plant tissue and soil samples were measured by the elementary analyzer (Vario EI cube, Elementar Analysensystem GmbH, Germany)-isotopic ratio mass spectrometer (DELTA V Advantage, Thermo Fisher Scientific, Inc., Bremen, Germany). $^{15}\text{N}\text{-NO}_3^-$ abundance in the outflow and soil media extract was measured by reducing NO_3^- to N_2O by the $\text{VCl}_3\text{-NaN}_3$ method [28]. $^{15}\text{N}\text{-TN}$ abundance in the outflow was determined by converting TN to NO_3^- by digestion with alkaline potassium persulfate and measuring the abundance of $^{15}\text{N}\text{-NO}_3^-$. $^{15}\text{N}\text{-NH}_4^+$ abundance in the outflow or soil media extract was measured by converting NH_4^+ to N_2O by NaBrO and NH_2OH in strongly acidic medium [17]. The $^{15}\text{N}\text{-N}_2\text{O}$ and $^{30}\text{N}_2$ abundances were directly measured by isotopic ratio mass spectrometer. The $^{15}\text{N}\text{-TN}$, $^{15}\text{N}\text{-NO}_3^-$, $^{15}\text{N}\text{-NH}_4^+$ and $^{15}\text{N}\text{-N}_2\text{O}$ were all reported in atom%: $^{15}\text{N}/(^{14}\text{N} + ^{15}\text{N}) \times 100\%$.

2.6. Data analysis

All statistical analysis was conducted by the R package v4.1.2. One-way ANOVA was conducted to analyze if plant species significantly affected N leaching, specific root surface area, mean root diameter, root biomass and ^{15}N immobilization, and it is assumed a significant difference when $p < 0.05$.

3. Results and discussion

3.1. Tracing of ^{15}N leached from BRSs

The average concentrations of effluent TN for BR-P, BR-A, BR-C were 1.50, 1.37 and 0.95 mg N L^{-1} , respectively, with small relative standard deviations in the range of 0.08 ~ 0.23 $\text{mg N}\cdot\text{L}^{-1}$ ($n = 5$) (see Fig. S1), indicating that the BRSs became stable.

3.1.1. ^{15}N abundance in the effluent

After ^{15}N -labeled N species enter BRSs, ^{15}N labels are leached from BRSs not only in the current rainfall-runoff event but also in the subsequent rainfall-runoff events. As shown in Fig. 3A, from the 1st event to the 5th event, the $^{15}\text{N}\text{-TN}$ leached from BR-P, BR-A and BR-C decreases from 8.96 to 0.24 mg event^{-1} , from 12.52 to 0.37 mg event^{-1} and from 4.92 to 0.14 mg event^{-1} , respectively. The abundances of ^{15}N in the outflow of these BRSs tend to be stable and approach to natural abundance in the 4th and 5th rainfall event, which is consistent with Wang's study that only used $^{15}\text{N}\text{-NH}_4^+$ labeling [30]. This suggests that immediate leaching and fast leaching mainly come from the current and previous four rainfall-runoffs, respectively.

3.1.2. The N species in the leached ^{15}N

Fig. 3B shows the proportions of N species in TN originated from ^{15}N -labeled influent in the consecutive rainfall-runoff events. From the 1st event to the 5th event, the proportions of NO_3^- originated from ^{15}N -labeled influent for BR-P, BR-A and BR-C decrease from 69.65% to 2.91%, from 62.12% to 1.29%, from 80.38% to 2.40%, respectively. Instead, from the 1st event to the 5th event, the proportions of DON in TN originated from ^{15}N -labeled influent for BR-P, BR-A and BR-C increase from 25.42% to 95.24%, from 3.58% to 95.09% and from 15.14% to 95.41%, respectively. However, the proportions of NH_4^+ in TN originated from ^{15}N -labeled influent for all BRSs have no obvious change trend (BR-P: range 0.26–4.93%; BR-A: range 3.61–34.29%; BR-C: range 2.69–10.63%). These results indicate that NO_3^- accounts for the largest

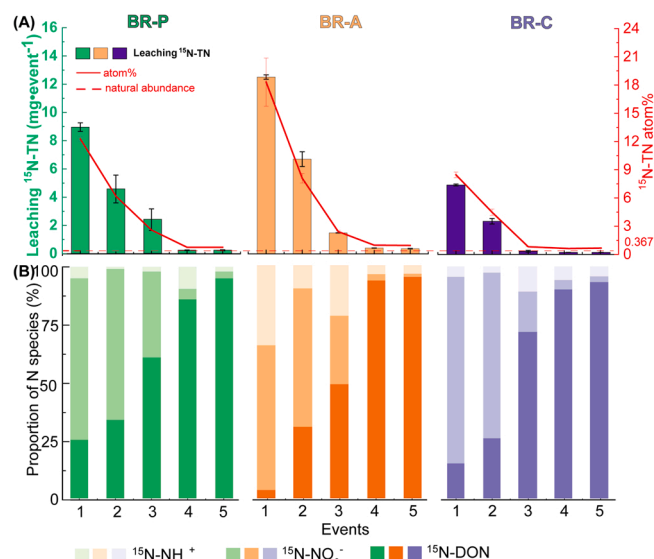


Fig. 3. Leaching ^{15}N -TN and its atom% (A), proportion of each N species (B) to TN originated from one-time ^{15}N -labeled influent in consecutive rainfall-runoff events. BR-P, BR-A and BR-C denote the bioretention systems with *Phragmites australis*, *Acorus calamus* L., *Cana indices* L., respectively. (mean $\pm$ s.d., $n = 3$).

proportion of ^{15}N -TN leached from these BRSs in the first two rainfall-runoff events, while DON accounts for the largest proportion in the latter three rainfall-runoff events, suggesting a fast transformation of inorganic to organic N.

3.2. Source apportionment of N leached from BRSs

3.2.1. Contribution of different sources to leached N

Fig. 4 A shows leaching TN and the percentages of immediate leaching, fast leaching and slow leaching to total leaching for BR-P, BR-A and BR-C. The amounts of immediate leaching, fast leaching and slow leaching for BR-P, BR-A and BR-C are in the range of 9.33 ~ 23.74 mg, 5.44 ~ 17.02 mg and 34.60 ~ 53.13 mg, respectively, and the percentages to total leaching are in the range of 17 ~ 32%, 10 ~ 22% and 46 ~ 73%, respectively. These results indicate that slow leaching accounts for the largest proportion among the three sources of N leached

from BRSs, followed by immediate leaching and fast leaching. Also, the ratios of immediate, fast and slow leaching to influent N loads as 2.1%, 1.7% and 6.6% for BR-P, respectively; 3.0%, 2.1% and 4.3% for BR-A, respectively; 1.1%, 0.7% and 4.8% for BR-C, respectively; indicating that the accumulated effect of slow N leaching is large.

3.2.2. The N species in leached N from different sources

The proportions of N species in TN leached from three sources are shown in Fig. 4B. The proportions of NH_4^+ in immediate leaching, fast leaching and slow leaching are in the range of 4.49~34.32%, 1.51~11.24% and 32.02~68.12%, respectively. The proportions of NO_3^- in immediate leaching, fast leaching and slow leaching are in the range of 62.09~80.37%, 49.43~59.54% and 8.28~15.49%, respectively. The proportions of DON in immediate leaching, fast leaching and slow leaching are in the range of 3.59~25.25%, 36.84~47.29% and 16.38~55.39%, respectively. These results indicate that NO_3^- accounts for the largest proportion of TN from immediate leaching and fast leaching, while DON and NH_4^+ mainly account for slow leaching.

3.2.3. Comparison between BRSs with different plant species

As shown in Fig. 4A, the total leaching TN of BR-C is the lowest among three types of BRSs because its immediate leaching and fast leaching are the lowest, and slow leaching is relatively low. Although there is no significant difference in total leaching between BR-P and BR-A, their source compositions are different: BR-A has more leaching TN from immediate leaching and fast leaching, while BR-P has more leaching TN from slow leaching.

3.3. Soil adsorption and immediate leaching

The main N species in immediate leaching is NO_3^- (62.09% ~ 80.37%) (Fig. 4B), which is mainly related to the poor adsorption of NO_3^- by soil media, although there is a saturated zone at the bottom of each BRS with ORPs ranging from -200.6 ~ -137.7 mv (Fig. S2) that could facilitate denitrification in this study. Instead, substantial NH_4^+ can be adsorbed by soil media due to cation exchange, which is advocated by many studies [3,27,30,32]. In addition, though influent DON in the form of urea can easily be hydrolyzed and then rapidly mineralized to NH_4^+ via microbial enzymes [14,25], complete hydrolysis of urea need to take several days even in the fertile loam [26]. In this study, 3.59~25.25% of DON detected in immediate leaching indicates that most of influent DON may also be retained by soil media with limited urea decomposition considering the low ^{15}N - NH_4^+ concentrations.

The immediate leaching of BR-A is the largest, followed by BR-P and BR-C, which could mainly be influenced by the morphology of plant roots (Fig. 5). *Acorus calamus* L. of BR-A and *Phragmites australis* of BR-P are featured by rhizome roots, while *Cana indices* L. of BR-C is featured by globular roots. The globular roots of *Cana indices* L. have the larger surface area that could have promoted N uptake [1], and the largest root diameter may have limited the occurrence of preferential flow [34], leading to the lowest immediate leaching in BR-C. Instead, for two plants with rhizome roots, there is more immediate leaching in BR-A than in BR-P, especially more NH_4^+ , suggesting stronger preferential flow, likely due to the larger biomass but smaller diameter of *Acorus calamus* L. roots than *Phragmites australis* roots.

Therefore, NO_3^- is the species most subject to immediate leaching, and immediate leaching relates to root morphology (biomass, surface area, diameter, etc.). Root morphology can influence plant uptake and preferential flow and thus, immediate leaching. Besides, we can infer that soil characteristics such as cation exchange capacity, porosity, and soil moisture should also influence immediate leaching by altering the adsorption capacities for NH_4^+ and DON and the retention of pore water with NO_3^- .

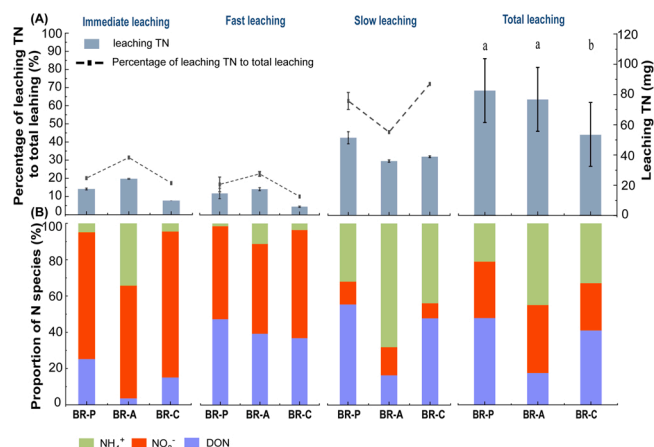


Fig. 4. Leaching TN and percentage of immediate leaching, fast leaching and slow leaching to total leaching (A), proportion of N species to TN from different sources (B) in three types of BRSs. Total leaching is the sum of immediate leaching, fast leaching and slow leaching. BR-P, BR-A and BR-C denote the bioretention systems with *Phragmites australis*, *Acorus calamus* L., *Cana indices* L., respectively. Different letters mean significant differences at $p < 0.05$. (mean $\pm$ s.d., $n = 3$).

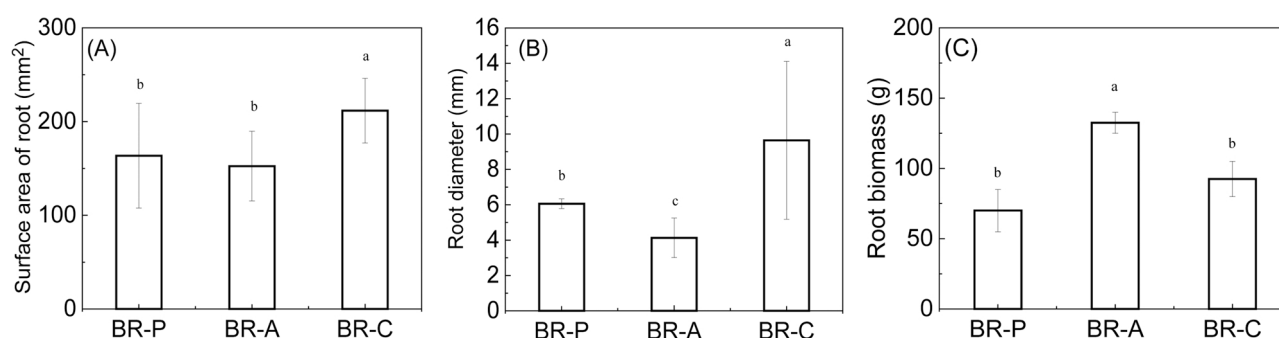


Fig. 5. Surface area of root (A), root diameter (B), and root biomass (C) for three types of BRSS. BR-P, BR-A and BR-C denote the bioretention systems with *Phragmites australis*, *Acorus calamus* L., *Canna indica* L., respectively. Different letters indicate significant differences at $p < 0.05$. (mean $\pm$ s.d., $n = 3$).

3.4. N transformation and fast leaching

The main N species of fast leaching for BRSS is NO_3^- (49.43% ~ 59.54%) (Fig. 4B), which can be mainly related to nitrification process during the dry periods [10], as evidenced by the positive net nitrification rates, especially in BR-P and BR-A (Table 1). Fast leaching of DON (36.84% ~ 47.29%) is also significant, and considering the limited amount of DON in immediate leaching, fast leaching of DON could occur after influent N species undergoes adsorption, desorption, building up in roots and microorganisms, and partial decomposition [15,20].

The fast leaching of BR-A is largest, followed by BR-P and BR-C (Fig. 4A), which could be mainly affected by N transformation. Firstly, the order of NO_3^- amount in fast leaching is $\text{BR-A} > \text{BR-P} > \text{BR-C}$, which is in line with the order of GN and NN (Table 1). The higher abundances of 16 S rRNA, nitrifying (AOA *amoA* and AOA *amoB*) and denitrifying (*nirK*, *narG* and *nosZ*) genes (Fig. 6), but significantly lower ORPs (57.4 ~ 336.1 mv) (Fig. S2) in BR-P and BR-A than in BR-C, indicate BR-P and BR-A are more dynamic in N cycling, especially in nitrification and denitrification, which could account for their more NO_3^- in fast leaching in BR-C. The most significant fast leaching of NO_3^- in BR-A is due to the medium abundances of nitrifying and denitrification genes as well as medium ORPs (83.9 ~ 119.2 mv), leading to the highest GN and NN. Secondly, the order of NH_4^+ amount in fast leaching, i.e. $\text{BR-A} > \text{BR-P} > \text{BR-C}$, is consistent with the order of GM and NM rates, suggesting the mineralized NH_4^+ may be more readily subject to fast leaching (Table 1). Instead, the lowest I_a rate in BR-C corresponds to the least fast leaching of DON, while the I_a and fast leaching of DON in BR-P and BR-A are similar, suggesting that NH_4^+ immobilization could be an important source of DON in fast leaching. It is possible that organic N directly from NH_4^+ immobilization is more labile and readily subject to fast leaching [11].

Therefore, NO_3^- and DON are the species most subject to fast leaching, and fast leaching mainly relates to N transformations. The nitrifying and denitrifying communities are widespread in bioretention systems

and the coupling of nitrification and denitrification determines the net accumulation of NO_3^- thus fast leaching of NO_3^- . Fast leaching of DON could relate to NH_4^+ immobilization processes. These findings point out the importance of N transformations in determining fast leaching, likely due to the products of N transformation are more readily subject to leaching [5,30]. This underscoring the potentials of microbial engineering practices to improve the N removal performances of bioretention systems.

3.5. N release and slow leaching

The main N species of slow leaching for BRSS are DON and NH_4^+ (Fig. 4B) [13,24]. The majority of influent N should eventually be immobilized into soil or assimilated by plants, and stored as organic N (Table S1), as disclosed in previous studies [6,30]. For example, *Phragmites australis*, *Acorus calamus* L. and *Canna indica* L. account for 13.07%, 14.91%, and 23.03% of influent ^{15}N , respectively; the soils of BR-P, BR-A and BR-C account for 63.30%, 47.99%, and 72.37% of influent ^{15}N , respectively (Fig. 7); the gaseous losses mainly via denitrification in BR-P, BR-A and BR-C account for 21.2%, 32.1% and 6.3%, respectively. Thus, most of DON and NH_4^+ in the slow leaching should be originated from the plant-soil ecotone, where soil organic matter is released by mineralization to NH_4^+ and converting “inert” species to DON. By comparing the N species composition in slow leaching with those in immediate and fast leaching, the amounts of NO_3^- are less, probably due to the lack of substantial NH_4^+ in soil for nitrification. Meanwhile, since there are interactions between soil, plant roots and microbial communities, we infer plant uptake should also influence slow leaching.

The slow leaching of BR-P is largest, followed by BR-C and BR-A (Fig. 4A), which can be related to plant uptake and soil retention. BR-P has the smallest plant uptake and relatively low soil retention accounting for the largest slow leaching, while BR-C has the largest plant uptake, largest soil retention, and relatively low slow leaching (Fig. 7), suggesting large plant uptake and soil retention can help withhold more N and prevent from slow leaching. In addition, Also, the retention capacity of N in soil can partly be determined by N transformation as shown in Table 1. For example, while DON is the most important N species in slow leaching in BR-P and BR-C, the slow leaching of NH_4^+ is significantly larger than that of DON in BR-A (Fig. 4), likely due to the rapid mineralization process in BR-A (Table 1).

Therefore, DON and NH_4^+ are the species most subject to slow leaching, and mainly originate from the release of soil organic matter. Plant uptake and soil retention can influence slow leaching, and soil retention can be determined by N transformation processes like mineralization to NH_4^+ or decomposition from “inert” species to labile DON. Slow leaching of DON and NH_4^+ can then occur in BRSS all the time, and N has a long-term legacy effect in BRSS. Plant selection, soil amelioration, and microbial control to improve N withholding capacities of plants and soils [20] could be promising measures to reduce slow leaching.

Table 1
N transformation rates in BRSS ($\text{mg N kg}^{-1} \text{ d}^{-1}$).

Bioretention systems	GM	NM	GN	NN	I_a
BR-P	8.48 ^b ± 0.73	0.22 ± 1.52	1.00 ^a ± 0.23	0.42 ^a ± 0.25	7.25 ^b ± 1.48
BR-A	14.32 ^a ± 4.85	0.35 ± 1.14	1.32 ^a ± 0.07	0.67 ^a ± 0.41	12.65 ^a ± 4.38
BR-C	3.56 ^c ± 0.74	0.08 ± 0.89	0.83 ^b ± 0.09	0.02 ^b ± 0.06	2.64 ^c ± 0.51

GM: gross mineralization rate; NM: net mineralization rate; GN: gross nitrification rate; NN: net nitrification rate; I_a : NH_4^+ immobilization rate; BR-P, BR-A and BR-C denote the bioretention systems with *Phragmites australis*, *Acorus calamus* L., *Canna indica* L., respectively. Different letters mean significant difference at $p < 0.05$. (mean $\pm$ s.d., $n = 12$).

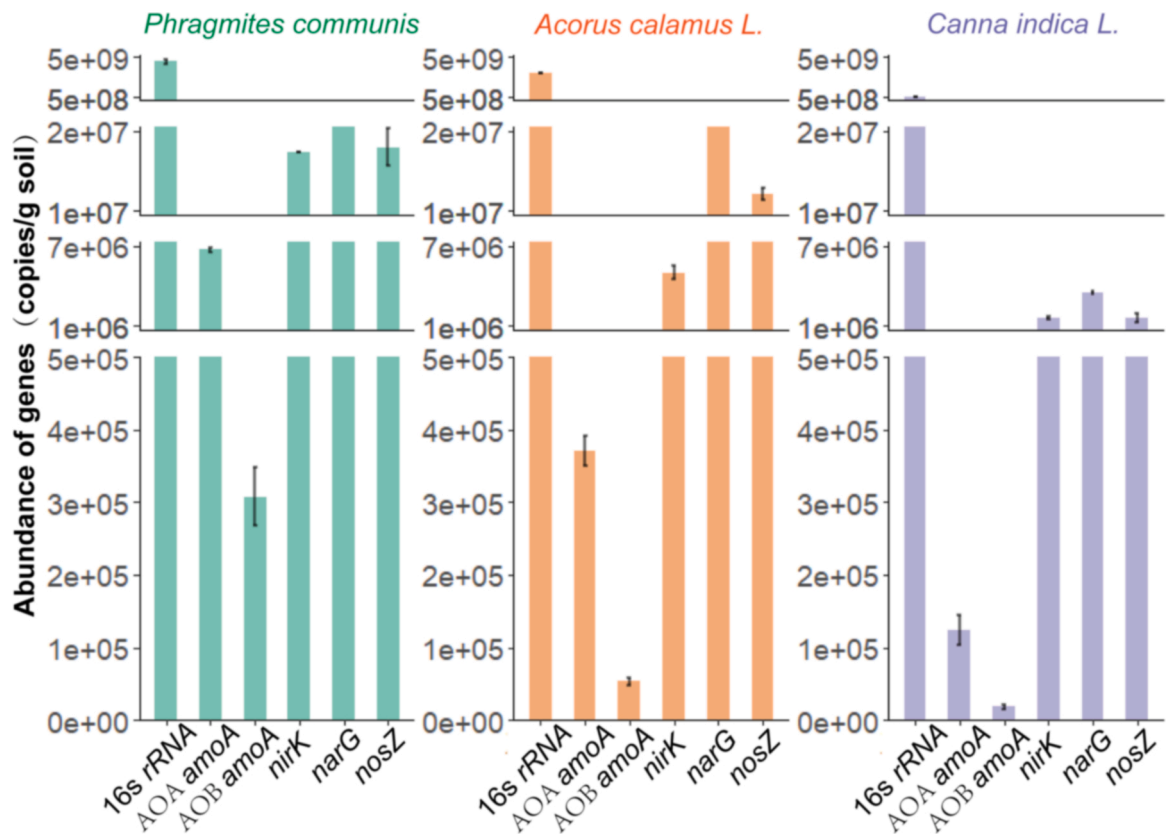


Fig. 6. Abundances of 16 S rRNA, nitrifying and denitrifying genes in bioretention systems with *Phragmites australis*, *Acorus calamus L.*, *Cana indices L.* (mean $\pm$ s.d., n = 3).

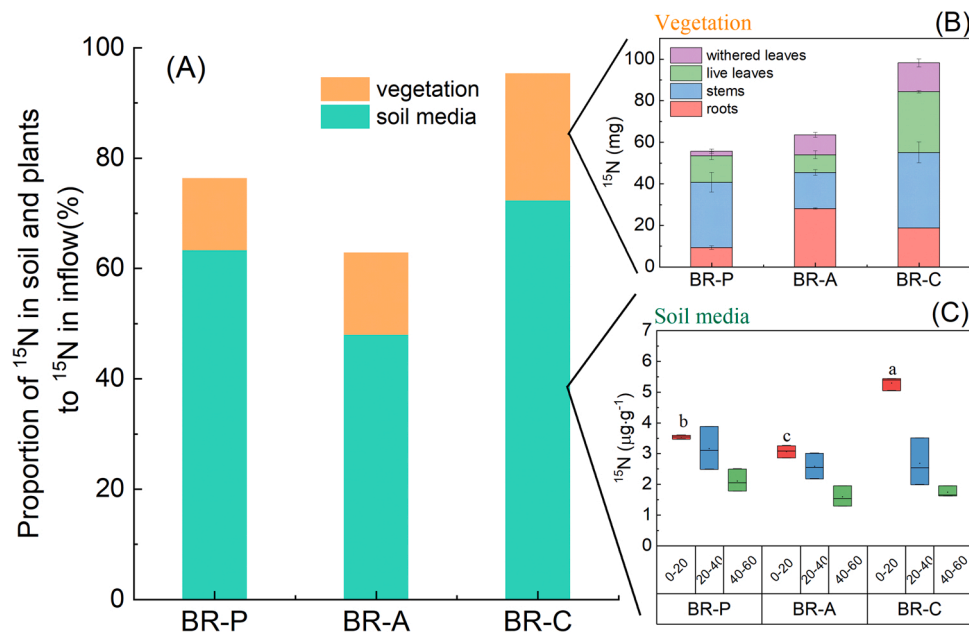


Fig. 7. Proportion of ^{15}N in soil and plants to ^{15}N in inflow (A), distribution of ^{15}N in plant tissues (B), and distribution of ^{15}N in different soil layers (C). BR-P, BR-A and BR-C denote the bioretention systems with *Phragmites australis*, *Acorus calamus L.*, *Canna indices L.*, respectively. Different letters indicate significant differences ($p < 0.05$).

4. Conclusions

This study has proposed the existences of three sources of N leaching, for the first time apportioned the three sources, and unraveled the

controlling mechanisms of each source of N leaching based on ^{15}N labeling technique. Immediate leaching and fast leaching mainly come from the current and previous several rainfall-runoffs (e.g. four events in this study), respectively, while the three sources contribute to total

leaching in the order of slow leaching > immediate leaching > fast leaching. NO_3^- is the dominant forms in immediate leaching, NO_3^- and DON in fast leaching, while DON and NH_4^+ in slow leaching.

Immediate leaching mainly relates to soil characteristics and root morphology (e.g. type, surface area and diameter) that can influence adsorption, preferential flow and plant uptake; fast leaching mainly relates to N transformation such as nitrification and immobilization during the dry periods that can influence the availabilities of NO_3^- and DON; slow leaching mainly relates to N release of organic matter from plant-soil ecotone via mineralization to NH_4^+ or decomposition from relatively “inert” organic N to DON. Source compositions are influenced by plant types: plants with large root surface area and diameter as well as high uptake rates can reduce immediate leaching, while plant-soil interactions may influence N transformations to impact fast leaching and slow leaching.

5. Implications and limitations

The differentiation of three sources of N leaching in BRSs and their controlling mechanisms are synthesized in Fig. 8. Influent N firstly undergoes soil adsorption and some N leaches out directly along the plant roots or soil pores, i.e. immediate leaching; then temporarily stored N leaches out after going through intense N transformation processes (e.g. mineralization, nitrification, denitrification and immobilization), i.e. fast leaching; finally, some N upheld by plants and retained by soil is transformed, released, and leaches out mainly in the form of DON and NH_4^+ , i.e. slow leaching.

With acquaintance of the source compositions and the controlling mechanisms, optimization of soil media, plant species and structure of BRSs can be well-directed and niche-targeting at different sources of N leaching. For rainfall-runoff with high NH_4^+ concentrations, fast leaching of NO_3^- and DON owing to nitrification and immobilization may occur, and the amendments of strong N-free adsorbents to withhold NH_4^+ and inhibitors from nitrification or immobilization should be reasonable. For rainfall-runoff with high NO_3^- concentrations, the selection of plants with NO_3^- preference and amendments of N-free anion adsorbents should minimize the immediate leaching of NO_3^- , while the saturated zone and augmentation of denitrifying microbes could promote the permanent removal of NO_3^- [16]. For rainfall-runoff with high DON concentrations, it is wise to amend N-free strong adsorbents to withhold DON and lower the diversity and abundance of microbes for mineralization and decomposition. However, in this study, there are still appreciable amounts of NO_3^- in immediate and fast leaching, though a saturated zone was set up for each BRS, suggesting the saturated zone may help remove NO_3^- but not entirely. Also, despite significant differences in each source of leaching, the differences in N leaching amount between BRSs with different plants are relatively small compared to the total leaching amount, suggesting the plant selection could influence N removal performances but marginally. Therefore, after overall evaluation, we advise that N-free mineral adsorbents to modify soil media of BRSs may be a low-cost but relatively effective approach to alleviate each source of N leaching.

This study is a successful demonstration of source apportionment of N leaching from BRSs in the column experiment, but up-scale generalizations in real BRS practices need to be made with cautions. In fact, N loads from rainfall-runoff events are variable, and how the source apportionment of N leaching can be feasible under different influent N loads in real BRS practices, remains unknown. Also, the length of ADP could pose significant effects on source apportionment of N leaching, and shorter ADPs would likely lead to larger contributions of immediate leaching in that the adsorption capacity of soil may become weaker with increased unprocessed N in soil. A dynamic analytical model of N leaching from BRSs to differentiate and test the changes of immediate leaching, fast leaching and slow leaching will also be desired in the future.

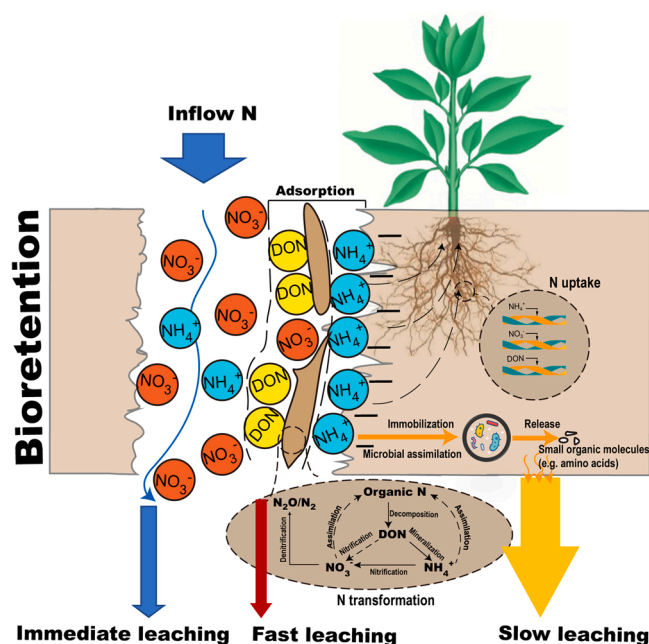


Fig. 8. (a) Diagram demonstrating the three sources of N leaching and their controlling mechanism in BRSs. (b) The thickness of the arrows represents the relative importance of each source of N leaching.

CRediT authorship contribution statement

Wei Ding, Fan Wang: Contributed equally to this paper, designed the study. **Wei Ding:** Performed the research, analyzed the data and interpreted the results. **Wei Ding, Fan Wang, Huapeng Qin, Chenxi Xia:** Contributed to the writing of the original manuscript. **Fan Wang, Huapeng Qin:** Contributed to the revision of the manuscript. All authors approve of the final article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, including effluent TN concentration, physical and chemical properties of soil during the experimental process, the changes of ORP during antecedent dry time and primer sequences for each functional gene used in this study, in the online version.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the

online version at [doi:10.1016/j.jece.2023.109819](https://doi.org/10.1016/j.jece.2023.109819).

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